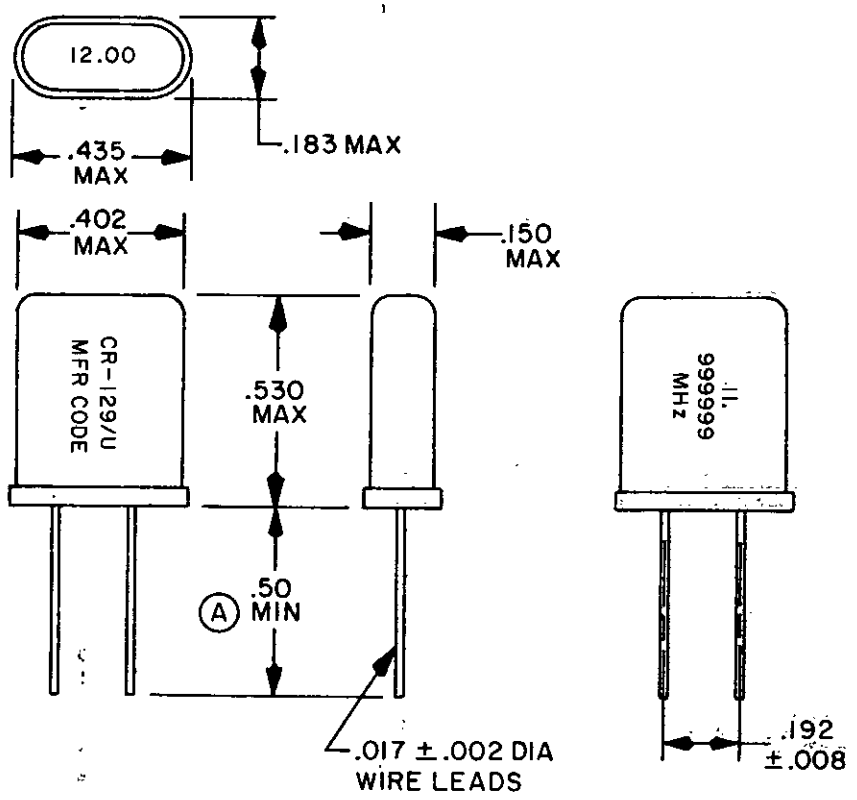
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-129/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 9 to 12 MHz; fundamental; noncontrolled; anti-resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-129/U.

Ⓐ denotes changes

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FSC 5955

Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	9.0 to 12.0 MHz, inclusive
Capacitance, shunt	7.0 pF, maximum
Mode of oscillation	Fundamental
Government reference standard test set	TS-330/TSM
Rated drive level	5.0 ±1.0 mW
Calibration values:	
Equivalent resistance	13 ohms
Crystal current	20 mA
Antiresonance, load capacitance	32.0 ±0.5 pF
Operating temperature range (noncontrolled)	-55°C to +90°C, inclusive
Frequency tolerance	±35 ppm
Equivalent resistance	MHz / Ohms, maximum 9.0 to 10.0 / 35 10.0+ to 12.0 / 25
Shock: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance	±15%, or 2 ohms, whichever is greater
Vibration: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15%, or 2 ohms, whichever is greater
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15%, or 2 ohms, whichever is greater
Aging:	
Frequency change permitted	±5 ppm

1/ Method 202 of MIL-STD-202; consisting of 18 impact shocks of 15 g's each having a duration of 11 milliseconds; 3 shocks along each direction in each of 3 mutually perpendicular planes.

2/ Method 204 of MIL-STD-202; test condition A

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 85
DLA - ES

User activities:
Air Force - 17, 19, 84

Agent:
DLA - ES

Preparing activity:
Army - ER

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